Short questions

|!|BCESB0080000001|!| ★

For each part of the breathing system listed in column I, select from column II one phrase that correctly describes it. Put the letter in the space provided. Some may require more than one answer. (5 marks)

Column I		***Column II***
Bronchus	(a)_____	A. Supported by complete rings of cartilage
Air sac	(b)_____	B. Supported by C-shaped rings of cartilage
Trachea	(c)_____	C. The site where gas exchange takes place
		D. Contains cilia on its inner surface

##

(a) A, D (1, 1)

(b) C (1)

(c) B, D (1, 1)

_____

(5 marks)

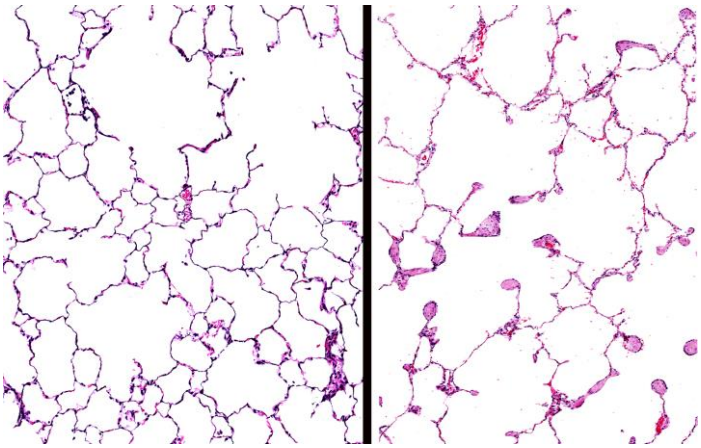
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	2

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|!|BCESB0080000002|!| ★

The photomicrographs below show sections through healthy lung tissue and lung tissue from a patient with emphysema. Both photomicrographs are of the same magnification.



Healthy lung tissue

Lung tissue from a patient with emphysema

(a) Give ***two*** differences that can be seen between the healthy lung tissue and the lung tissue from the patient with emphysema. (2 marks)

(b) Explain why emphysema patients may find it difficult to climb stairs. (3 marks)

##

(a) The number of air sacs in the lung tissue from emphysema patient is lower. (1)
The walls of some air sacs become thickened. (1)

(b) In a patient suffering from emphysema, the total surface area of the air sacs is reduced. (1)
The rate of gas exchange / oxygen uptake becomes slower. (1)
The patient needs to pant / breathe harder and faster to take in enough oxygen. (1)

(5 marks)

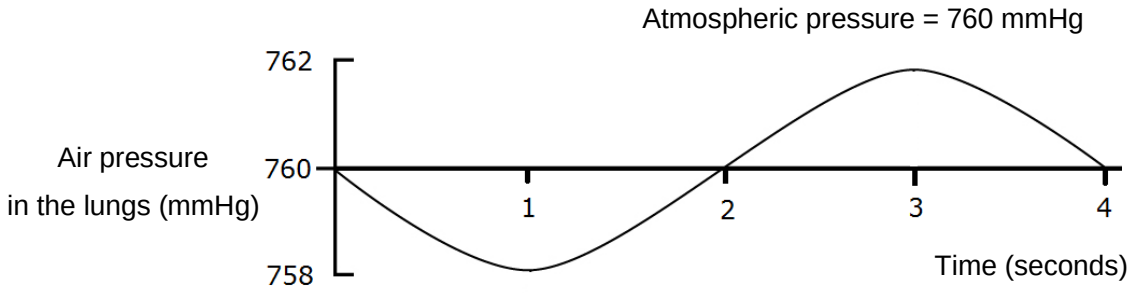
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##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	3

|!|BCESB0080000003|!| ★

The graph below shows the changes in pressure inside the lungs of a healthy person who breathes normally.



(a) Calculate the rate of breathing in breaths per minute. Show your working.

(2 marks)

(b) In which time period is this person breathing out? Explain you answer with reference to the graph. (2 marks)

##

(a) One complete breath takes 4 seconds.

60 ÷ 4 (1)

= 15 breaths per minute (1)

(b) From 2 to 4 seconds (1)

It is because during this time period, the air pressure in the lungs is higher than atmospheric pressure. (1)

(4 marks)

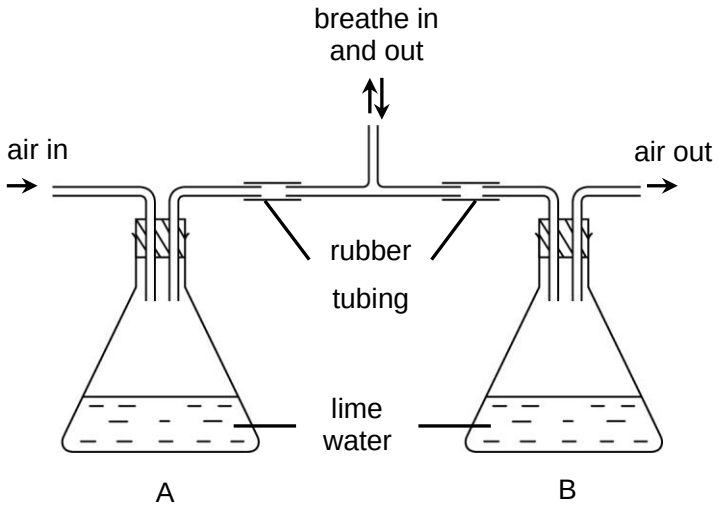
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	Skills assessed	Marks
(a)	Mathematic skills	2
(b)	Analysing data	1
	Applying concepts	1

|!|BCESB0080000004|!| ★ **SI**

The diagram below shows an experimental set-up used by a student to compare the composition of air breathed in and out. However, the set-up could not work.



(a) Explain why the experimental set-up would not work. (1 mark)

(b) In the diagram above, make changes to the experiment set-up so that it will work. (2 marks)

(c) Suppose the changes in (b) have been made. What change could be observed in the lime water in flasks A and B after several breaths? (2 marks)

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(d) What can be deduced from (c) concerning the difference in gas composition between the air breathed in and breathed out by the student? (1 mark)

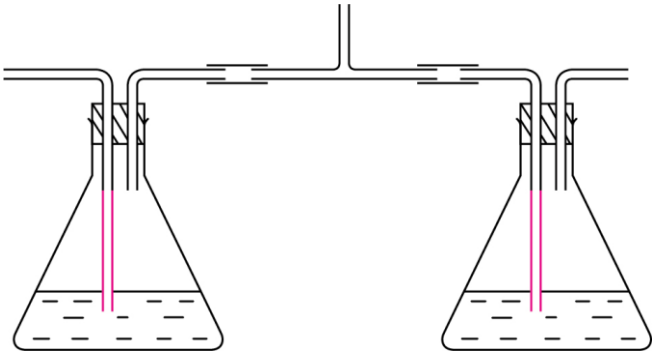
_____

_____

##

(a) When the student breathed in or out, air did not pass through the lime water. (1)

(b) Correct drawing: elongate the left glass tube in each conical flask so that the tube is immersed in the lime water (2)



(c) Flask A: remains clear (1)

Flask B: becomes cloudy (1)

(d) The air breathed out has a higher carbon dioxide content than the air breathed in. (1)

_____

(6 marks)

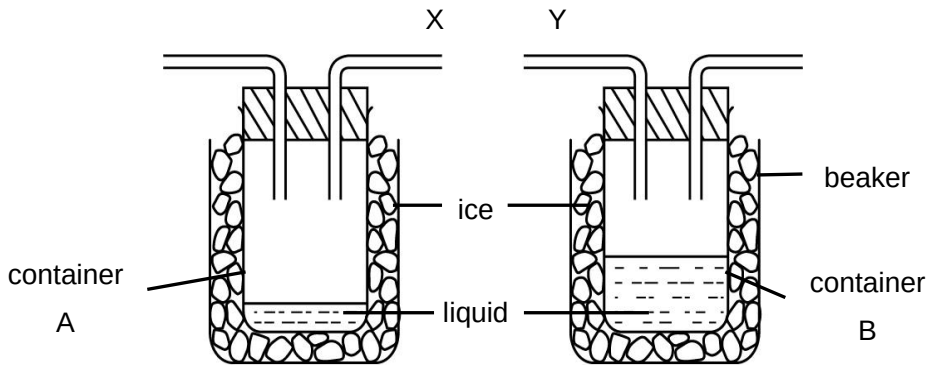
##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Practical skills	2
(c)	Applying concepts	2
(d)	Applying concepts	1

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|!|BCESB0080000005|!| ★ **SI**

The diagram below shows the apparatus used to compare the composition of exhaled and inhaled air. A student breathed in through glass tube X and out through glass tube Y for 15 minutes. At the end of the experiment, a liquid was collected inside both containers and formed a layer at the bottom.



(a) What is the name of the liquid collected? How can you test the liquid? (3 marks)

(b) With reference to the experimental results, what conclusion can be drawn from this experiment? (2 marks)

##

(a) The liquid is water. (1)
Test the liquid with a piece of anhydrous cobalt chloride paper. (1)
If the liquid is water, the test paper will turn from blue to pink. (1)

(b) The amount of liquid (water) collected in container B is larger than that in container A. (1)
Exhaled air contains more water vapour than inhaled air. (1)

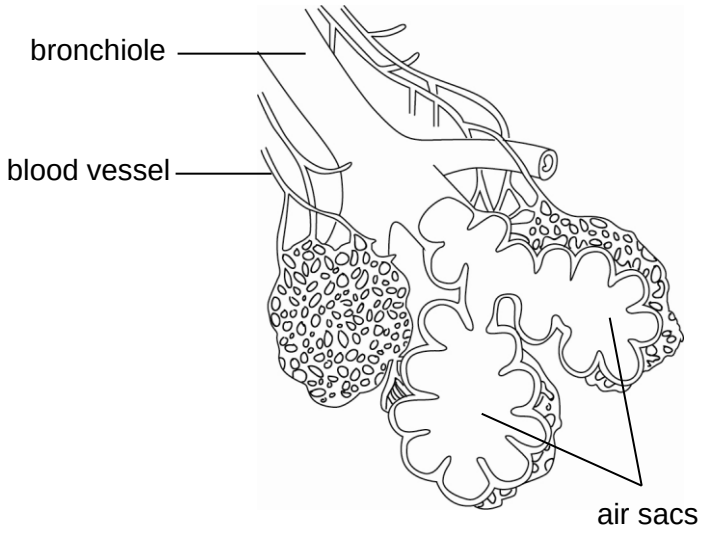
(5 marks)

##

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	Skills assessed	Marks
(a)	Practical skills	3
(b)	Analysing data	2

|!|BCESC0080000006|!|
A human lung has about 80 million air sacs. The following diagram shows some air sacs in a human lung.



(a) Give ***two*** features of the air sacs that allow large amounts of oxygen to enter the blood. (2 marks)

_____

_____

(b) Describe how oxygen passes from the air into the blood. (2 marks)

_____

_____

(c) Inhalation helps bring fresh air that contains oxygen to the lungs. Describe how the actions of intercostal muscles and diaphragm muscles achieve inhalation. (2 marks)

_____

_____

_____

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##

(a) Any ***two*** of the following: (2)

- Air sacs have a large surface area. (1)
- The walls of air sacs are thin. (1)
- Air sacs are surrounded by a dense network of blood capillaries. (1)

(b) Oxygen dissolves in the water on the inner surface of air sacs. (1)

Oxygen then diffuses across the walls of air sac and capillaries into the blood. (1)

(c) Intercostal muscles contract. (1)

Diaphragm muscles contract. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

|!|BCESB0080000007|!| ★

Our respiratory tract has structures that can filter and clean the inhaled air.

(a) State ***three*** adaptations of our respiratory tracts to filter and clean the inhaled air.

(3 marks)

(b) In mining sites, the environment is full of dust. The dust particles cannot be effectively filtered and removed by the respiratory tract of the miners. State and explain how the dusty environment affects the gas exchange in the miners.

(3 marks)

##

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(a) Nasal hairs in the nostrils filter out large dust particles. (1)
Mucus-secreting cells secrete mucus which traps bacteria and dust particles. (1)
Beating action of cilia of ciliated epithelial cells moves the mucus towards the pharynx. (1)

(b) Dust particles deposit on the inner wall of the air sacs of lungs. (1)
This increases the diffusion distance of gases, (1)
reducing the efficiency of gas exchange. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Applying concepts	3

|!|BCESC0080000008|!|
Exhaled air has a different composition when compared with inhaled air.

(a) State ***two*** differences in the composition between inhaled air and exhaled air. (2 marks)

(b) Explain the differences you stated in (a). (2 marks)

##

(a) Any ***two*** of the following: (2)
- Exhaled air contains less oxygen. (1)
- Exhaled air contains more carbon dioxide. (1)
- Exhaled air contains more water vapour. (1)

(b) Any ***two*** of the following: (2)
- Oxygen in inhaled air diffuses from the air sacs into the blood. (1)
- Carbon dioxide, produced from cellular respiration, diffuses from the blood into the air sacs. (1)
- Water evaporates from the moist respiratory tract and the air sacs of lungs. (1)

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(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2

|!|BCESB0080000009|!| ★

The following photomicrograph shows some red blood cells.



(a) What is the site of the production of red blood cells? (1 mark)

_____

(b) With reference to the photomicrograph, state the shape of a red blood cell and how its shape can help in the transport of oxygen. (2 marks)

_____

_____

_____

(c) Besides the shape, state and explain ***two*** other adaptations of the red blood cells to the transport of oxygen. (4 marks)

_____

_____

_____

_____

##

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(a) (Red) bone marrow (1)

(b) The red blood cells are biconcave disc-shaped. (1)

This shape can increase the surface area to volume ratio for oxygen to diffuse into and out of the cells more efficiently. (1)

(c) Red blood cells are packed with haemoglobin molecules. (1)

Haemoglobin can combine with oxygen and act as oxygen carrier. (1)

Mature red blood cells do not have a nucleus. (1)

This provides more space to hold more haemoglobin molecules. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	4

|!|BCESC0080000010|!|

Fill in the blanks to complete the following paragraph. (6 marks)

The movement of air into and out of the lungs is called ventilation. During inhalation, the diaphragm muscles (a)_____, causing the diaphragm to become (b)_____. At the same time, the intercostal muscles contract to (c)_____ the rib cage. The volume of the thoracic cavity (d)_____. The air pressure inside the lungs becomes (e)_____ than the atmospheric pressure. As a result, air moves (f)_____ the lungs.

##

(a) contract (1)

(b) flattened (1)

(c) raise (1)

(d) increases (1)

(e) lower (1)

(f) into (1)

(6 marks)

##

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	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

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